E-WASTE: WHAT HAPPENS TO OLD PHONES AND COMPUTERS?

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INTRODUCTION

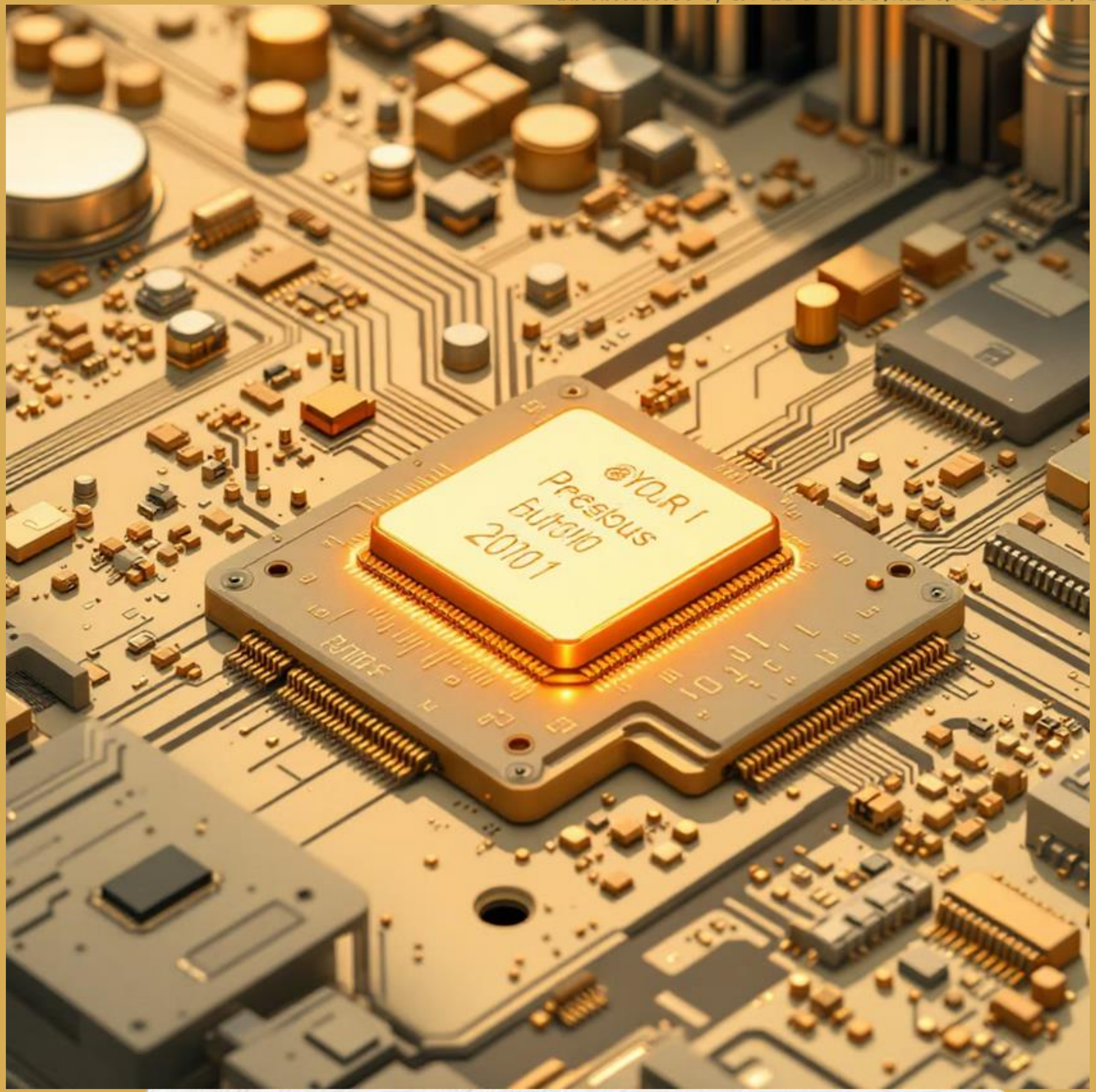
What is E-Waste?

DEFINITION

Electronic waste (e-waste) refers to any electrical or electronic device that has been discarded, including both working and broken items that are disposed of or intended for reuse, resale, salvage, recycling, or disposal. This encompasses a vast array of consumer electronics, from the smallest mobile phone components to large household appliances.

KEY CHARACTERISTICS

- Contains valuable materials including gold, silver, copper, and rare earth elements that can be recovered through proper recycling
- Generated from diverse sources including consumer electronics, information technology equipment, and household appliances
- Contains hazardous substances such as lead, mercury, cadmium, and brominated flame retardants
- Requires specialised handling and processing due to complex material composition



Simple Rule: If anything plugs in or requires a charge, it becomes part of the e-waste stream when discarded.

TYPES OF E-WASTE

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Electronic waste encompasses a diverse range of products, each presenting unique challenges and opportunities for recycling and resource recovery. Understanding these categories is crucial for developing targeted management strategies.

Information Technology Equipment Waste (ITEW)

Desktop computers, laptops, servers, mainframes, and associated peripherals including keyboards, mice, and monitors. This category represents the fastest-growing segment due to rapid technological advancement and shorter shelf lives.

Consumer Electronics & Entertainment Waste (CEEW)

Televisions, music systems, gaming consoles, digital cameras, and home entertainment systems. These devices often contain valuable precious metals and rare earth elements alongside hazardous materials.

Large Scale Equipment Waste (LSEEW)

Washing machines, refrigerators, air conditioners, and other major household appliances. These typically contain significant amounts of steel, copper, and aluminium, making them valuable for material recovery.

Small Electronics Waste (EETW)

Mobile phones, calculators, radios, and portable electronic devices. Though individually small, these items collectively represent a significant source of precious metals recovery.

Temperature Exchange Equipment Waste (TLSEW)

Cooling and freezing equipment including refrigerators, air conditioners, and heat pumps. These require careful handling due to refrigerant gases and insulation materials.

Medical Device Waste (MDW)

Medical equipment, diagnostic devices, and healthcare electronics requiring specialised handling due to potential biological contamination and sensitive data concerns.

Lighting Equipment Waste (LIW)

LED lights, fluorescent lamps, and specialty lighting systems containing mercury and other hazardous materials requiring careful processing.

E-WASTE STATISTICS IN INDIA

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Alarming Growth Trajectory

India's e-waste generation has reached unprecedented levels, reflecting the nation's rapid digital transformation and increasing consumer electronics adoption. The country generated **1.6 million metric tonnes** of e-waste in FY 2022, representing **more than double** the amount generated just four years earlier in 2018.

This exponential growth places India as the **third-largest generator** of e-waste globally, trailing only China and the United States. The primary drivers include rapid digitisation initiatives, increasing smartphone penetration, shorter device replacement cycles, and growing affluence leading to higher electronics consumption.

> The rapid pace of technological advancement coupled with declining device costs has created a perfect storm for e-waste generation in India.

Critical Recycling Gap

Despite the massive generation of e-waste, India faces a significant infrastructure and processing challenge. Only **6% of e-waste is properly recycled** through authorised facilities that meet environmental and safety standards.

The remaining **95% of recycling activities** occur through the informal sector, where workers often lack proper safety equipment and training. This informal processing frequently involves hazardous practices such as acid baths for precious metal recovery and open burning of cables, leading to severe environmental and health consequences.

The lack of proper collection infrastructure, limited awareness among consumers about proper disposal methods, and insufficient authorised recycling capacity contribute to this alarming recycling gap.

1.6M
Metric Tonnes
E-waste generated in India (FY 2022)

3rd
Global Ranking
India's position in worldwide e-waste generation

6%
Proper Recycling
Percentage processed through authorised facilities

95%
Informal Sector
Percentage handled by unauthorised recyclers

ENVIRONMENTAL IMPACT

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The environmental consequences of improper e-waste disposal represent one of the most pressing environmental challenges of the digital age. Electronic devices contain a complex mixture of valuable and hazardous materials that, when not properly managed, can cause severe and long-lasting environmental damage.

HAZARDOUS SUBSTANCES IN E-WASTE

Lead

Found in CRT monitors, circuit boards, and soldering materials. Causes severe damage to the nervous system, blood formation processes, and kidney function. Particularly harmful to children's cognitive development.

Mercury

Present in fluorescent lamps, LCD screens, and batteries. Causes chronic damage to the brain, central nervous system, and respiratory system. Bioaccumulates in food chains, particularly affecting aquatic ecosystems.

Cadmium

Used in rechargeable batteries, pigments, and stabilisers. Accumulates in kidneys and causes neural damage, bone weakness, and has been linked to various forms of cancer.

Chromium

Found in metal housing and corrosion protection systems. Causes skin diseases, allergic reactions, and lung cancer. Hexavalent chromium is particularly carcinogenic and mutagenic.

Brominated Flame Retardants

Used in plastic housings and circuit boards for fire prevention. Disrupts endocrine system function, affects thyroid hormone regulation, and can cause developmental disorders.

ENVIRONMENTAL CONSEQUENCES

SOIL CONTAMINATION

Heavy metals and toxic chemicals leach from improperly disposed e-waste into soil systems, making agricultural land infertile and contaminating food crops grown in affected areas.

AIR POLLUTION

Uncontrolled burning of e-waste releases harmful gases including dioxins, furans, and particulate matter, contributing to air quality degradation and respiratory health problems.

WATER POLLUTION

Toxic substances migrate through groundwater systems, contaminating drinking water supplies and affecting entire aquatic ecosystems through bioaccumulation in fish and other marine life.

RESOURCE DEPLETION

Valuable metals including gold, silver, and rare earth elements are permanently lost when e-waste ends up in landfills, necessitating additional mining activities that cause further environmental destruction.

E-WASTE RECYCLING PROCESS

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COLLECTION & TRANSPORTATION

E-waste is gathered through dedicated recycling bins, manufacturer take-back programmes, and collection drives. Specialised vehicles equipped with proper containment systems transport materials to authorised processing facilities. This stage requires careful handling to prevent damage and contamination during transit.

SORTING & DISMANTLING

Trained technicians manually sort devices by type and condition. Components are carefully dismantled using appropriate tools to separate different materials. This labour-intensive process requires expertise to identify valuable components whilst safely removing hazardous substances such as batteries and mercury-containing parts.

SHREDDING & SEPARATION

Mechanical breakdown occurs through industrial shredders that reduce devices to small fragments. Magnetic separation techniques extract ferrous metals, whilst water separation and air classification methods separate materials based on density differences. Advanced optical sorting technologies identify specific plastic types and metal compositions.

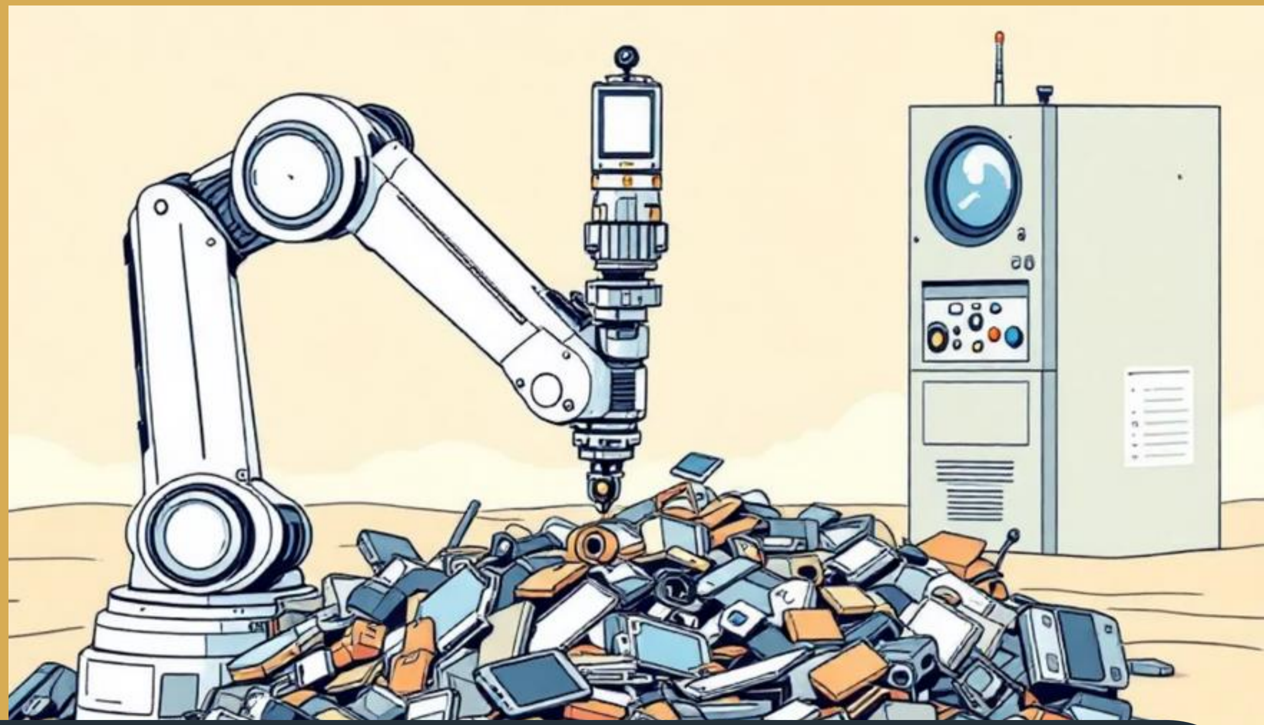
RESOURCE RECOVERY

Sophisticated chemical and physical processes recover precious metals including gold, silver, copper, and platinum from circuit boards and components. Plastics undergo reprocessing for manufacturing new products. Hazardous materials receive appropriate treatment or secure disposal through certified hazardous waste management facilities.

TECHNOLOGY INTEGRATION

Modern recycling facilities increasingly employ artificial intelligence and robotics to improve sorting accuracy and worker safety. Automated systems can identify specific component types and material compositions with greater precision than manual sorting, leading to higher recovery rates and reduced contamination.

Advanced hydrometallurgical processes allow for the recovery of rare earth elements that were previously considered economically unviable to extract, making recycling operations more profitable and environmentally beneficial.



GOVERNMENT REGULATIONS

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E-Waste Management Rules 2022
The most recent comprehensive guidelines providing updated definitions, expanded scope, and enhanced enforcement mechanisms. These rules supersede previous regulations and incorporate lessons learned from international best practices and domestic implementation experiences.

Central Pollution Control Board (CPCB)
National-level implementation authority responsible for policy development, standards setting, and coordination with state agencies. CPCB maintains databases of authorised recyclers and monitors compliance across the country.

State Pollution Control Boards (SPCB)
State-level monitoring and enforcement agencies that issue authorisations, conduct inspections, and ensure compliance with environmental standards. SPCBs work directly with local recycling facilities and waste generators.

EXTENDED PRODUCER RESPONSIBILITY (EPR)

The EPR framework places legal responsibility on manufacturers for the entire lifecycle of their electronic products, from design and production through to end-of-life management. This approach incentivises manufacturers to design more sustainable products and invest in recycling infrastructure.

- **Collection Obligations:** Manufacturers must establish collection systems for their products at end-of-life
- **Recycling Targets:** Specific percentage targets for material recovery and recycling
- **Registration Requirements:** Mandatory registration with pollution control boards
- **Annual Reporting:** Detailed reporting on collection, recycling, and disposal activities

CASE STUDY: GREEN INDIA RECYCLING

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Green India Recycling represents a successful model of compliant, environmentally responsible e-waste management in India. This case study demonstrates how proper regulatory compliance, technological investment, and environmental stewardship can create a sustainable business model whilst addressing the nation's e-waste challenges.

Company Profile

ESTABLISHMENT
Founded in **2015** in Thane, Maharashtra, strategically located near major urban centres to facilitate efficient collection and processing operations.

CERTIFICATIONS
MPCB-certified (Maharashtra Pollution Control Board) and **ISO 14001:2015** certified, demonstrating commitment to environmental management standards.

PROCESSING CAPACITY
Authorised to process **3,000 metric tonnes annually** of various e-waste categories, with additional capacity for lead-acid battery recycling.

SERVICES OFFERED

Collection & Segregation

Comprehensive collection services including doorstep pickup for corporate clients, segregation of different e-waste types, and secure data destruction services with certified data wiping procedures.

Environmentally Sound Recycling

State-of-the-art recycling processes that maximise material recovery whilst minimising environmental impact, including precious metal recovery and plastic reprocessing capabilities.

Dismantling Operations

Systematic dismantling of electronic devices using appropriate tools and techniques, careful separation of hazardous components, and preparation of materials for further processing stages.

Specialised Battery Recycling

Dedicated lead-acid battery recycling facility with 1,800 metric tonnes annual capacity, recovering lead, plastic, and acid components for reuse in new battery manufacturing.

CONCLUSION

CRISIS
E-waste is a major and growing environmental issue in India and globally.

HAZARDS
Most e-waste is recycled informally, causing significant health and ecological risks.

SOLUTIONS
Strict regulation, technological innovation, and public awareness are essential for effective e-waste management.

MODELS
Successful examples like Green India Recycling prove that sustainable business models can benefit both the environment and the economy.

ACTION
Immediate action and collective effort are needed to bridge the recycling gap and mitigate the negative impacts of e-waste.

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Central Pollution Control Board (CPCB) and State Pollution Control Boards (SPCB), E-Waste Management Rules 2022: Framework for environmentally sound management of electronic waste and compliance requirements in India.

Green India Recycling Case Study

Green India E-Waste Re-Cycling OPC Pvt. Ltd., “Company Profile, Certifications, and E-Waste Recycling Process,” Green India Recycling PPT, 2024: Overview of MPCB-certified e-waste management and battery recycling practices.

Statistical Data on E-Waste in India

India’s e-waste generation has more than doubled in four years to 1.6 million metric tonnes in FY 2022, ranking 3rd globally, with only 6% being properly recycled through the authorized sector.

ISO Standards for Biodegradability of Plastics

Bureau of Indian Standards, IS ISO 14851:1999, IS ISO 14852:1999, IS ISO 15985:2004, etc.: Methods for measuring biodegradability of plastics in environmental conditions as per Schedule-I of Indian Rules.

Lead-Acid Battery and Hazardous Waste Authorisation

Maharashtra Pollution Control Board, “Grant or Renewal of Authorisation by State Pollution Control Board for Disposal Facilities,” Lead-Batteries-Authorisation.pdf, 2024: Authorisation guidelines and compliance for recycling hazardous lead-acid batteries.